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A Hybrid Deep Learning Model for Cross-Project Software Defect Prediction

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Slide 1: Introduction

- Software defects reduce reliability and increase maintenance cost
- Traditional models fail to generalize across projects
- Within-Project Prediction gives misleading accuracy
- **Cross-Project Defect Prediction (CPDP)** is more realistic
- Goal: Predict defects in unseen projects

Slide 2: Problem Statement

- Class imbalance (more non-defective than defective modules)
- Domain shift between projects
- Poor generalization of existing models
- High F1 but low real predictive power (biased models)

Slide 3: Objectives

- Improve cross-project defect prediction
- Handle class imbalance effectively
- Build a generalized deep learning model
- Evaluate using **MCC, AUC, F1-score**
- Compare performance across datasets

Slide 4: Innovation Type

New + Improved Solution

- Hybrid architecture:
  - Multi-scale CNN
  - BiLSTM
  - Multi-head Attention
- Key innovations:
  - Focal Loss (handles imbalance)
  - Dynamic class weights
  - Threshold tuning for best F1
  - Cross-project validation

Slide 5: Methodology

- Load the dataset and preprocess the data (convert labels, normalize values, and prepare input format)
- Train the model on multiple projects and test it on a different unseen project (cross-project approach)
- Evaluate performance using MCC, AUC, and F1-score
- Use a hybrid deep learning model combining CNN (for feature extraction), BiLSTM (for sequence understanding), and Attention (to focus on important features)
- Train the model using Adam optimizer, apply focal loss to handle imbalance, and use early stopping to avoid overfitting

Slide 6: Demo

https://colab.research.google.com/drive/1YEGh5zaR4uGCVcegC8bXx2q9BvzK0Rcg?usp=sharing

Slide 7: Results and Key Findings

Model	Hybrid (CNN-LSTM-Dense)			Hybrid (CNN-Transformer)			Hybrid (Multi-scale CNN + BiLSTM + Multi-head Attention)		
Metric	FM	MCC	AUC	FM	MCC	AUC	FM	MCC	AUC
LC	0.8948	0.3211	0.751	0.8948	0.3217	0.7655	0.4368	0.4695	0.7889
ML	0.8259	0.1765	0.7381	0.828	0.1733	0.7537	0.2804	0.219	0.6286
PDE	0.8337	0.3104	0.747	0.8382	0.3062	0.7402	0.2517	0.2093	0.7155
EQ	0.5288	0.2192	0.5561	0.5695	0.2982	0.6749	0.1958	0.2613	0.8247
JDT	0.8174	0.4326	0.825	0.798	0.3758	0.7798	0.4803	0.4389	0.8194
Mean	0.7801	0.292	0.7234	0.7841	0.2931	0.7372	0.329	0.3196	0.7554

Model	Hybrid (CNN-LSTM-Dense)			Hybrid (CNN-Transformer)			Hybrid (Multi-scale CNN + BiLSTM + Multi-head Attention)		
Metric	FM	MCC	AUC	FM	MCC	AUC	FM	MCC	AUC
Safe	0.7193	0.4217	0.7941	0.7609	0.5027	0.8048	0.7404	0.446	0.774
Zxing	0.6543	0.155	0.6208	0.6196	0.0932	0.6201	0.6667	0.5027	0.7594
Apache	0.6773	0.37	0.6707	0.7004	0.4029	0.7298	0.4609	0.0615	0.588
Mean	0.6836	0.3156	0.6952	0.6937	0.3329	0.7183	0.6227	0.3367	0.7071

- **Evaluation Metrics**
  - **MCC (Primary Metric)** – robust for imbalanced data
  - **AUC** – measures ranking ability
  - **F1-score** – balance of precision & recall
- **Key Observations**
  - Achieves **higher MCC than baseline models**
  - More Balanced prediction
  - Handles class imbalance better.

Slide 8: Achievements

- Model implementation completed
- Experimental evaluation on AEEEM and ReLink datasets
- Achieved improved MCC performance
- Research paper writing in progress
- Implemented cross-project validation framework

Slide 9: Expected Outcomes

- Better defect prediction across projects
- Improved generalization on unseen data
- Helps developers identify high-risk modules
- Potential for research publication
- Scope for future work (transfer learning, domain adaptation)

Slide 14: References

- AEEEM Dataset (D’Ambros et al.)
- ReLink Dataset (Wu et al.)
- ROC Analysis – Fawcett
- MCC – Matthews

Thank You